Practice Lab:

DSE Analytics, Machine Learning, Apriori Algorithm

- This Practice Lab is dependent on Discussion Unit 7602, where most of the objects we create in this lab were introduced.
- This Practice Lab is also dependent on Discussion Unit 7548/7549, where we learn how to compile, set up a Spark/Scala project structure, other.

- This Practice Lab requires a working DSE system, with DSE Analytics enabled.

- Because of the dependency on the "dse spark-submit" utility, this Practice Lab requires a ssh(C) prompt on at least one node operating DSE Analytics.

Challenge 1: Prerequisites

Prerequisites:

- Instructions are provided for Linux
- We will operate below recommended settings; 1 OS node only, minimum 8 GB RAM, 12 GB RAM preferred

- That you have an operating single node DSE Analytics cluster.
- All work done as ‘root’

- You need to complete Practice Lab 7549 first, because we are going to re-use its project filesystem structure, other.

Working Goal:

```
From: Maury_Atwater
To: DSE_HOTSHOT

Subject: Need this now !!!

We need to start upselling ! CSV file; 10k
orders, 43k items sold, 170 unique items,
4.4 items per transaction.

We need a set of association rules to
apply on the Web site !

-MA
```

Maury Atwater, President of Atwater's

Challenge 1: Deliver Apriori

- You are given the CSV source data file. Place this file in an easily spelled absolute pathname.

- You are given the single Scala application source file, which is the Apriori application.
- Let's save steps and time, and re-use the infrastructure from the 7548/7549 Practice Lab.

- Save that original App.scala file to another location.
- Overwrite the contents of App.scala with the Apriori source code.

Challenge 1: Compile and Run

From the my-app folder:

- Compile via, mvn package
- Run via,

    dse spark-submit --class com.datastax.enablement.bootcamp.App target/my-app-1.0.jar

Challenge 2 (Optional): Same for Customer Churn

- Save App.scala
- Replace contents of App.scala with the single Customer Churn application code

Challenge 3 (Optional): Clean Up

- Using the instructions from Practice Lab 7549, actually instantiate two new project parent directories, so that we are not (cheating, and overwriting the contents of App.scala)

- Compile and run all 3 programs; the original and the two new

Practice Lab:

Lessons Learned

End of Unit: